

# DA 54. Analysis of changes to the SPGZ in 2025 in terms of grounds

Zelimkhan Gudiev

2025-12-17

## Contents

|                                                      |    |
|------------------------------------------------------|----|
| 0. Install packages and attach libraries             | 3  |
| 1. Business task, context and scope of work          | 4  |
| 1.1. Key tasks and deliverable(s)                    | 4  |
| 1.1.1. Business task                                 | 4  |
| 1.1.2. Key stakeholders                              | 4  |
| 1.1.3. SMART questions                               | 4  |
| 1.1.4. Scope of work (SOW)                           | 4  |
| 1.2 Guiding questions                                | 5  |
| 2. Data preparation                                  | 6  |
| 2.1. Key tasks and deliverables                      | 6  |
| 2.1.1. Collecting the necessary data                 | 6  |
| 2.1.2. Downloading and storing data                  | 6  |
| 2.1.3. Data organization                             | 6  |
| 2.1.4. Data credibility determination                | 6  |
| ROCCC framework                                      | 6  |
| Data integrity                                       | 6  |
| 2.1.5. Bias                                          | 7  |
| 2.1.6. Conclusion on the admissibility of using data | 7  |
| 2.2 Guiding questions                                | 7  |
| 3. Data processing                                   | 8  |
| 4. Data analysis                                     | 15 |
| 4.1 Key tasks and deliverable(s)                     | 15 |
| 4.1.1. Data aggregation                              | 15 |
| 4.1.2. Data organizing and formatting                | 15 |
| 4.1.3. Performing calculations                       | 15 |
| 4.1.4. Identifying trends and relationships          | 15 |
| 4.1.5. Summary of the analysis                       | 15 |
| 4.2 Guiding questions                                | 15 |
| 5. Report                                            | 16 |
| 5.1 Key tasks and deliverable(s)                     | 16 |
| 5.1.1. Determining communication approach            | 16 |
| 5.1.2. Data visualizations creating                  | 16 |
| 5.1.3. Presenting findings                           | 16 |
| 5.1.4. Ensuring accessibility                        | 16 |

|                                                             |    |
|-------------------------------------------------------------|----|
| 5.1.5. Support visualizations and key findings . . . . .    | 16 |
| 5.2 Guiding questions . . . . .                             | 16 |
| 6. Usage results . . . . .                                  | 17 |
| 6.1. Key tasks and deliverable(s) . . . . .                 | 17 |
| 6.1.1. Top three recomendations based on analysis . . . . . | 17 |
| 6.2 Guiding questions . . . . .                             | 17 |

## 0. Install packages and attach libraries

```
# install.packages("psych")
# install.packages("lubridate")
# install.packages("purrr")
# install.packages("GGally")
# install.packages("skimr")
# install.packages("Writexl")
# install.packages("corrplot")
library(psych)
library(google sheets4)
library(tidyr)
library(dplyr)
library(janitor)
library(ggplot2)
library(lubridate)
library(purrr)
library(GGally)
library(readxl)
library(skimr)
library(writexl)
library(corrplot)
```

# 1. Business task, context and scope of work

## 1.1. Key tasks and deliverable(s)

### 1.1.1. Business task

Determine the number of changes made to the SPGZ in 2025 for each of possible scenarios (cases) for the subsequent determination of frequency of scenarios (cases) in which customers receive notifications (in accordance with “EAIST\_JB-8765 /, ”) and scenarios (cases) that entail the need for customers to make changes to the lot specifications.

#### Notes

- The assignment to carry out of this task was issued at the “Meet-145 , ( )” meeting.
- This task is carrying out in the card “DA-54 / / 2025 ”
- The results of this task will be used in the following tasks:
  - SGL-105 ( )
  - Meet-145 , ( )

### 1.1.2. Key stakeholders

The key stakeholders of this task are following:

- Department of Economic Policy and Development of the City of Moscow
- Department of Competition Policy of the City of Moscow
- Moscow city customers

### 1.1.3. SMART questions

The result of completing this task should be the answers to the following questions

| N | Type                                            | Question                                                                                                                                          |
|---|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | - Specific - Measurable - Relevant - Time-bound | How often did the SPGZ change in 2025 for each of the scenarios (cases) described in “ ” file, prepared during the work on task “IS_Task-318 , ”. |
| 2 | ...                                             | ...                                                                                                                                               |

### 1.1.4. Scope of work (SOW)

| N | Deliverables                                                              | TimelineMilestones                                            | Results                                                                   | What SMART questions will be answered |
|---|---------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------|
| 1 | Dataset with information about changes SPGZ in 2025 from Catalog database | 18.12.2025Completion of extracting data from Catalog database | Dataset with information about changes SPGZ in 2025 from Catalog database | -                                     |
| 2 | Clean dataset                                                             | 19.12.2025Dataset is clean and usable                         | -                                                                         |                                       |

| N | Deliverables                                   | Timeline   | Milestones                                                    | Results                                                                                       | What SMART questions will be answered                                                                                                           |
|---|------------------------------------------------|------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Exploratory data analysis (EDA)                | 21.12.2025 | Completion of EDA - understanding of dataset is formed        | EDA highlights room for investigation. Questions were formulated based on the results of EDA. | -                                                                                                                                               |
| 4 | Engineered features in the form of new columns | 22.12.2025 | New columns engineered to provide more accessible information |                                                                                               |                                                                                                                                                 |
| 5 | Statistical insights                           | 24.12.2025 | Completion of statistical analysis                            | Statistical analysis conducted                                                                | How often did the SPGZ change in 2025 for each of the scenarios (cases) described in “ ” file, prepared during the work on task “IS_Task-318 ”. |
| 6 | Visual aids                                    | 25.12.2025 | Completion of visualizations                                  | Visualizations created                                                                        | -                                                                                                                                               |
| 7 | Final report                                   | 27.12.2025 | Final report submission                                       | Final report completed                                                                        | -                                                                                                                                               |

## 1.2 Guiding questions

| N | Question                                                             | Answer                                                                                                                                                                  |
|---|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | What problem need to be solved using the result of this task         | The problem that needs to be solved is that customers need to make changes in they lots if the SPGZ positions used in those lot’s specifications are changed or deleted |
| 2 | How can the insights (results) of this task drive business decisions | This analysis insights will be used to determine necessary EAIST upgrades and their priorities.                                                                         |

## 2. Data preparation

### 2.1. Key tasks and deliverables

#### 2.1.1. Collecting the necessary data

Data was collected from several Catalog database tables using the SQL query “spgz\_kpgz\_relations\_packages.sql”.

#### 2.1.2. Downloading and storing data

The data was downloaded and stored at the enterprise cloud and available is at this link

```
# df <- read.csv("C:/Users/GudievZK/Nextcloud/Data analysis/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_db_ktlg.xlsx")
# df_src <- read.csv("C:/Users/GudievZK/Nextcloud/Data analysis/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_db_ktlg.xlsx")

df <- read_excel("/Users/zelimkhan/Desktop/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_db_ktlg.xlsx", guess_max_col = 1000)
df_src <- read_excel("/Users/zelimkhan/Desktop/DA-54/2025.12.18_spgz_kpgz_ktru_pack_rels_db_ktlg.xlsx", guess_max_col = 1000)
```

#### 2.1.3. Data organization

The data is organized as a table (in csv format) containing 58682 rows. The rows contain information about versions of SPGZ positions which were changed or deleted in 2025.

#### 2.1.4. Data credibility determination

The ROCCC framework and Data integrity methodology were used to determine the data credibility

ROCCC framework

| N | ROCCC Dimension | Description                                                                                                                                                               | Conclusion                                                               | Comments                                    |
|---|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------|
| 1 | Reliable        | Source is dependable and consistent.                                                                                                                                      | See to the conclusion below the table                                    | It depends on ROCCC dimensions listed below |
| 2 | Original        | The data obtained from the original source (or if the data is not obtained from the original source, it is possible compere and verify the data with the original source) | The data obtained from the original source (Catalog database).           |                                             |
| 3 | Comprehensive   | Covers all necessary cases and variables                                                                                                                                  | The data covers all necessary cases and variables                        |                                             |
| 4 | Current         | The data source aren't out of date or irrelevant                                                                                                                          | The data is current and contains the necessary information for 2025 year |                                             |
| 5 | Cited           | The data source is cited and vetted                                                                                                                                       | The data source is cited and vetted                                      |                                             |

Conclusion on reliability of the data source based on the on ROCCC framework: The data source is reliable.

Data integrity

| N | Data integrity dimension | Description                                                                                                                                                                                    | Conclusion                                              |
|---|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1 | Accuracy                 | The degree to which the data conforms to the actual entity being measured or described                                                                                                         | The data conforms actual changes in the SPGZ in 2025    |
| 2 | Completeness             | Data completeness is a dimension of data quality that measures whether all necessary or required data is present and available for dataset to fulfill to intended purpose.                     | The dataset contains all necessary cases and variables. |
| 3 | Consistency              | Data follows the same formats and rules.                                                                                                                                                       | The data is consistency                                 |
| 4 | Trustworthiness          | The trustworthiness refers to the degree to which information can be relied upon as accurate, credible and secure, allowing users confidently make decisions and conduct analysis based on it. | The data is trustworthiness                             |

Conclusion on credibility of the data source based on Data integrity methodology: The source is credibility.

#### 2.1.5. Bias

The dataset come from Catalog database and reflect actual changes in the SPGZ in 2025

#### 2.1.6. Conclusion on the admissibility of using data

The data is credible and can be use for analysis

### 2.2 Guiding questions

| N | Questions                                                                    | Answers                                                                                                                                                             |
|---|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Where the data is located?                                                   | The data was downloaded from Catalog database and stored in the enterprise cloud at this link (see section “2.1.2. Downloading and storing data”)                   |
| 2 | How is the data organized?                                                   | The data is organized as a table consisting of 58682 and 47                                                                                                         |
| 3 | Are there issues with bias or credibility in the data? Does the data ROCCC?  | The data is unbiased and credible. Data is integrity and meets the ROCCC methodology requirements (see section the “2.1.4. Data credibility determination” section) |
| 4 | How are issues of licencing, privacy, security, and accessibility addressed? |                                                                                                                                                                     |

### 3. Data processing

Stage 1 of EDA: Data Discovery & Profiling

Table overview

```
df

## # A tibble: 58,682 x 47
##   TOTAL_ROWS ID_ENTITY_SPGZ_EAIST ID_SPGZ_EAIST ID_ENTITY_SPGZ_KATALOG
##   <chr>      <chr>          <chr>      <chr>
## 1 58682     10008020         1530772545 10008020
## 2 58682     10008020         1592957700 10008020
## 3 58682     10008215         1530772547 10008215
## 4 58682     10008215         1592956858 10008215
## 5 58682     10044442         10044442    10044442
## 6 58682     10044442         1640044123 10044442
## 7 58682     10052745         1530785272 10052745
## 8 58682     10052745         1575153004 10052745
## 9 58682     10052764         1530785269 10052764
## 10 58682    10052764         1575153001 10052764
## # i 58,672 more rows
## # i 43 more variables: ID_VERSION_SPGZ_KATALOG <chr>, PARENTID_SPGZ <chr>,
## #   CODE_KPGZ_SPGZ <lgl>, NAME_SPGZ <chr>, ISSTANDARDIZED_SPGZ <chr>,
## #   ISPROJECT_SPGZ <chr>, ISDELETED_SPGZ <chr>, ISEXCLUDED_SPGZ <chr>,
## #   ISEMIAS_SPGZ <chr>, SPGZATTRIBUTES_SPGZ <chr>, ADDITIONALUNIT_SPGZ <chr>,
## #   ISPCP_SPGZ <chr>, ATTRIBUTETARIF_SPGZ <chr>, KTRUID_SPGZ <chr>,
## #   OKPDID_SPGZ <chr>, ISKPGZ_SPGZ <chr>, CREATEDATE_SPGZ <chr>, ...
```

Clean names

```
# toString(names(df))
df <- clean_names(df)
```

Columns names

```
names(df)

## [1] "total_rows"          "id_entity_spgz_eaist"
## [3] "id_spgz_eaist"       "id_entity_spgz_katalog"
## [5] "id_version_spgz_katalog" "parentid_spgz"
## [7] "code_kpgz_spgz"      "name_spgz"
## [9] "isstandardized_spgz" "isproject_spgz"
## [11] "isdeleted_spgz"      "isexcluded_spgz"
## [13] "isemias_spgz"        "spgzattributes_spgz"
## [15] "additionalunit_spgz" "ispcp_spgz"
## [17] "attributetarif_spgz" "ktruid_spgz"
## [19] "okpdid_spgz"         "iskpgz_spgz"
## [21] "createdate_spgz"     "editdate_spgz"
## [23] "begindate_version_spgz" "enddate_version_spgz"
## [25] "is_kpgz"             "okpd2_id_spgz"
## [27] "okpd2_code_spgz"     "okpd2_name_spgz"
## [29] "okpd2_canceledate_spgz" "okpd2_isactual_spgz"
## [31] "okpd2_okpd2id_spgz"  "ktru_code_spgz"
## [33] "ktru_name_spgz"      "rels_old_pgz_id"
## [35] "rels_linktype"       "rels_new_pgz_id"
## [37] "rels_is_actual"      "rels_previous_pgz_id"
```

```
## [39] "rels_source_pgz_id"      "package_name"
## [41] "pgzpack_pgz_id"         "package_reason_id"
## [43] "package_user_name"      "package_reason_name"
## [45] "id_entity_kpgz"         "name_kpgz"
## [47] "code_kpgz"
```

Number of rows and columns

```
dim(df)
```

```
## [1] 58682  47
```

Data types

```
glimpse(df)
```

```
## Rows: 58,682
## Columns: 47
## $ total_rows      <chr> "58682", "58682", "58682", "58682", "58682", "~
## $ id_entity_spgz_eaist <chr> "10008020", "10008020", "10008215", "10008215"~
## $ id_spgz_eaist    <chr> "1530772545", "1592957700", "1530772547", "159~
## $ id_entity_spgz_katalog <chr> "10008020", "10008020", "10008215", "10008215"~
## $ id_version_spgz_katalog <chr> "30040471", "30384089", "30040473", "30384100"~
## $ parentid_spgz    <chr> "3206701", "3206701", "3206701", "3206701", "6~
## $ code_kpgz_spgz   <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ name_spgz        <chr> " ~
## $ isstandardized_spgz <chr> "1", "1", "1", "1", "0", "0", "0", "0", "0", "~
## $ isproject_spgz    <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ isdeleted_spgz    <chr> "0", "1", "0", "1", "0", "1", "0", "1", "0", "~
## $ isexcluded_spgz   <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ isemias_spgz      <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ spgzattributes_spgz <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ additionalunit_spgz <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ ispcp_spgz        <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ attributetarif_spgz <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ ktruid_spgz       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ okpdid_spgz       <chr> "39566", "39566", "39566", "39566", "44818", "~
## $ iskpgz_spgz       <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ createdate_spgz   <chr> "2024-08-26 18:10:59.000", "2025-03-19 15:41:3~
## $ editdate_spgz     <chr> "2024-08-26 16:08:59.092", "2025-03-19 15:07:3~
## $ begindate_version_spgz <chr> "2024-08-26 00:00:00.000", "2025-03-19 00:00:0~
## $ enddate_version_spgz <chr> "2025-03-19 00:00:00.000", NA, "2025-03-19 00:~
## $ is_kpgz          <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "~
## $ okpd2_id_spgz     <chr> "1750486", "2063125", "1750486", "2063125", "1~
## $ okpd2_code_spgz   <chr> "28.99.39.190", "28.99.39.190", "28.99.39.190"~
## $ okpd2_name_spgz   <chr> " , ~
## $ okpd2_canceldate_spgz <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ okpd2_isactual_spgz <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "~
## $ okpd2_okpd2id_spgz <chr> "8888313", "8888313", "8888313", "8888313", "8~
## $ ktru_code_spgz    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ ktru_name_spgz    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_old_pgz_id   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_linktype     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_new_pgz_id   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_is_actual    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ rels_previous_pgz_id <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
```

```
## $ rels_source_pgz_id    <chr> "10050054", "10050054", "10050055", "10050055"~
## $ package_name          <chr> " 17", "  ~
## $ pgzpack_pgz_id       <chr> "30040471", "30384089", "30040473", "30384100"~
## $ package_reason_id     <dbl> 5, 5, 5, 5, NA, 3, 5, 5, 5, 5, 5, 5, 5, 5, ~
## $ package_user_name     <chr> " ", " ~
## $ package_reason_name   <chr> " ", " ", " ~
## $ id_entity_kpgz        <dbl> 3206701, 3206701, 3206701, 3206701, 68912, 689~
## $ name_kpgz             <chr> " ", " ~
## $ code_kpgz             <chr> "01.25.93.12", "01.25.93.12", "01.25.93.12", "~
```

Data classes

```
sapply(df, class)
```

```
##          total_rows  id_entity_spgz_eaist      id_spgz_eaist
##          "character"      "character"      "character"
## id_entity_spgz_katalog id_version_spgz_katalog      parentid_spgz
##          "character"      "character"      "character"
##          code_kpgz_spgz      name_spgz      isstandardized_spgz
##          "logical"      "character"      "character"
##          isproject_spgz      isdeleted_spgz      isexcluded_spgz
##          "character"      "character"      "character"
##          isemias_spgz      spgzattributes_spgz      additionalunit_spgz
##          "character"      "character"      "character"
##          ispcp_spgz      attributetarif_spgz      ktruid_spgz
##          "character"      "character"      "character"
##          okpdid_spgz      iskpgz_spgz      createdate_spgz
##          "character"      "character"      "character"
##          editdate_spgz      begindate_version_spgz      enddate_version_spgz
##          "character"      "character"      "character"
##          is_kpgz      okpd2_id_spgz      okpd2_code_spgz
##          "character"      "character"      "character"
##          okpd2_name_spgz      okpd2_canceldate_spgz      okpd2_isactual_spgz
##          "character"      "logical"      "character"
##          okpd2_okpd2id_spgz      ktru_code_spgz      ktru_name_spgz
##          "character"      "character"      "character"
##          rels_old_pgz_id      rels_linktype      rels_new_pgz_id
##          "character"      "numeric"      "character"
##          rels_is_actual      rels_previous_pgz_id      rels_source_pgz_id
##          "character"      "logical"      "character"
##          package_name      pgzpack_pgz_id      package_reason_id
##          "character"      "character"      "numeric"
##          package_user_name      package_reason_name      id_entity_kpgz
##          "character"      "character"      "numeric"
##          name_kpgz      code_kpgz
##          "character"      "character"
```

Checking duplicates. Number of duplicate columns in each column.

```
sapply(df, function(x) sum(duplicated(x)))
```

```
##          total_rows  id_entity_spgz_eaist      id_spgz_eaist
##          58681      41066      17368
## id_entity_spgz_katalog id_version_spgz_katalog      parentid_spgz
##          41114      16702      56644
##          code_kpgz_spgz      name_spgz      isstandardized_spgz
```

```
##          58681          45324          58680
## isproject_spgz isdeleted_spgz isexcluded_spgz
##          58681          58680          58680
## isemias_spgz spgzattributes_spgz additionalunit_spgz
##          58681          58680          58680
## ispcp_spgz atributetarif_spgz ktruid_spgz
##          58679          58681          49485
## okpdid_spgz iskpgz_spgz createdate_spgz
##          58132          58681          37490
## editdate_spgz begindate_version_spgz enddate_version_spgz
##          49115          57627          55366
## is_kpgz okpd2_id_spgz okpd2_code_spgz
##          58681          54003          56779
## okpd2_name_spgz okpd2_canceldate_spgz okpd2_isactual_spgz
##          56786          58681          58680
## okpd2_okpd2id_spgz ktru_code_spgz ktru_name_spgz
##          56779          55178          57733
## rels_old_pgz_id rels_linktype rels_new_pgz_id
##          45371          58678          57115
## rels_is_actual rels_previous_pgz_id rels_source_pgz_id
##          58679          58681          32038
## package_name pgzpack_pgz_id package_reason_id
##          54331          17298          58676
## package_user_name package_reason_name id_entity_kpgz
##          58618          58676          56644
## name_kpgz code_kpgz
##          56644          56644
```

Checking uniqueness.

Number of unique rows.

```
df %>% n_distinct()
```

```
## [1] 58544
```

Number of unique values in each column

```
sapply(df, function(x) sum(n_distinct(x)))
```

```
##          total_rows id_entity_spgz_eaist id_spgz_eaist
##              1          17616          41314
## id_entity_spgz_katalog id_version_spgz_katalog parentid_spgz
##          17568          41980          2038
## code_kpgz_spgz name_spgz isstandardized_spgz
##              1          13358              2
## isproject_spgz isdeleted_spgz isexcluded_spgz
##              1              2              2
## isemias_spgz spgzattributes_spgz additionalunit_spgz
##              1              2              2
## ispcp_spgz atributetarif_spgz ktruid_spgz
##              3              1          9197
## okpdid_spgz iskpgz_spgz createdate_spgz
##              550              1          21192
## editdate_spgz begindate_version_spgz enddate_version_spgz
##              9567          1055          3316
## is_kpgz okpd2_id_spgz okpd2_code_spgz
```

```
##          1          4679          1903
##      okpd2_name_spgz okpd2_canceldate_spgz okpd2_isactual_spgz
##          1896          1          2
##      okpd2_okpd2id_spgz      ktru_code_spgz      ktru_name_spgz
##          1903          3504          949
##      rels_old_pgz_id      rels_linktype      rels_new_pgz_id
##          13311          4          1567
##      rels_is_actual      rels_previous_pgz_id      rels_source_pgz_id
##          3          1          26644
##      package_name      pgzpack_pgz_id      package_reason_id
##          4351          41384          6
##      package_user_name      package_reason_name      id_entity_kpgz
##          64          6          2038
##      name_kpgz      code_kpgz
##          2038          2038
```

Checking completeness

```
colSums(is.na(df))
```

```
##      total_rows id_entity_spgz_eaist id_spgz_eaist
##          0          75          61
##      id_entity_spgz_katalog id_version_spgz_katalog      parentid_spgz
##          0          0          0
##      code_kpgz_spgz      name_spgz      isstandardized_spgz
##          58682          0          0
##      isproject_spgz      isdeleted_spgz      isexcluded_spgz
##          0          0          0
##      isemias_spgz      spgzattributes_spgz      additionalunit_spgz
##          0          0          58681
##      ispcp_spgz      attributetarif_spgz      ktruid_spgz
##          1633          0          42773
##      okpdid_spgz      iskpgz_spgz      createdate_spgz
##          49819          0          0
##      editdate_spgz      begindate_version_spgz      enddate_version_spgz
##          1779          0          19829
##      is_kpgz      okpd2_id_spgz      okpd2_code_spgz
##          0          0          0
##      okpd2_name_spgz okpd2_canceldate_spgz okpd2_isactual_spgz
##          0          58682          0
##      okpd2_okpd2id_spgz      ktru_code_spgz      ktru_name_spgz
##          0          47586          47587
##      rels_old_pgz_id      rels_linktype      rels_new_pgz_id
##          40655          40655          40655
##      rels_is_actual      rels_previous_pgz_id      rels_source_pgz_id
##          40655          58682          22851
##      package_name      pgzpack_pgz_id      package_reason_id
##          613          613          613
##      package_user_name      package_reason_name      id_entity_kpgz
##          3638          613          0
##      name_kpgz      code_kpgz
##          0          0
```

Converting data types Converting all columns with fewer than 10 unique (non-NA) values into factors

```
# df <- df %>%
# mutate(
#   across(
#     where(~ n_distinct(.x, na.rm = TRUE) < 10),
#     as.factor
#   )
# )
```

Converting variables with date and time values from character format to date format.

```
df <- df %>%
  mutate(
    across(
      contains("date_") & where(is.character),
      ~ as.Date(ymd_hms(.x))
    )
  )
```

Filtering data

```
df <- df %>%
  filter(
    (isexcluded_spgz == 0 & !is.na(package_user_name)) & # deleting excluded spgz and spgz without author (which wh
    (
      year(editdate_spgz) == 2025 |
      year(enddate_version_spgz) == 2025)
    )
  )
```

Deleting duplicate rows

```
df <- df %>%
  distinct()
```

Deleting unnecessary columns

```
df <- df %>%
  select(id_entity_spgz_katalog, id_version_spgz_katalog, name_spgz, isstandardized_spgz, isproject_spgz, isdeleted_sp
  ispcp_spgz, createdate_spgz, editdate_spgz, begindate_version_spgz, enddate_version_spgz, okpd2_code_spgz,
  okpd2_name_spgz, ktru_code_spgz, ktru_name_spgz, rels_old_pgz_id, rels_linktype, rels_new_pgz_id, rels_is_actu
  rels_source_pgz_id, package_name, package_user_name, package_reason_name, id_entity_kpgz, name_kpgz, code_k

# toString(names(df))
```

```
dups <- df %>%
  group_by(across(everything())) %>% # group by all columns
  filter(n() > 1) %>% # keep groups with more than one row
  ungroup()
```

Adding new columns Adding a column with number of values for each id\_entity\_spgz\_katalog

```
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    n_id_entity_spgz_katalog = n()) %>%
  ungroup()
```

Adding a column with the numbers of unique values id\_version\_spgz\_katalog per id\_entity\_spgz\_katalog

```
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    n_unique_id_version_cataloge = n_distinct(id_version_spgz_katalog, na.rm = TRUE)) %>%
  ungroup()
```

Adding column with difference between “id\_entity\_spgz\_katalog” and “id\_version\_spgz\_katalog”

```
df <- df %>%
  mutate(
    diff_id_entity_id_version = n_id_entity_spgz_katalog - n_unique_id_version_cataloge
  )
```

Adding a column with deletion information

```
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    is_deleted_spgz = as.integer(sum(as.integer(isdeleted_spgz), na.rm = TRUE) > 1)
  ) %>%
  ungroup()
```

Adding column with information about relation...

Adding column with information about standardizing in 2025

```
df <- df %>%
  group_by(id_entity_spgz_katalog) %>%
  mutate(
    was_standartized_25 = as.integer(
      n_distinct(trimws(as.character(isstandardized_spgz)), na.rm = TRUE) > 1
    )
  ) %>%
  ungroup()
```

dssd

```
# rels_old_pgz_id <- df %>% filter(!is.na(rels_old_pgz_id)) %>% select(rels_old_pgz_id)
# df_rels <- df %>% filter(id_version_spgz_katalog %in% rels_old_pgz_id)

# df_0 %>% filter(
#   id_version_spgz_katalog == "30425543" |
#   rels_old_pgz_id == "30425543" |
#   id_entity_spgz_katalog == "29270927" |
#   id_entity_spgz_katalog == "30424361" |
#   id_version_spgz_katalog == "30425542"
# ) %>% # write_xlsx("/Users/zelimkhan/Desktop/DA-54/2025.12.21_df_0.xlsx")
# # write_xlsx("C:/Users/GudievZK/Nextcloud/Data analysis/DA-54/df_0.xlsx")
```

## 4. Data analysis

### 4.1 Key tasks and deliverable(s)

4.1.1. Data aggregation

4.1.2. Data organizing and formatting

4.1.3. Performing calculations

4.1.4. Identifying trends and relationships

4.1.5. Summary of the analysis

### 4.2 Guiding questions

| N | Questions                                     | Answers |
|---|-----------------------------------------------|---------|
| 1 | How should data be organized for analysis?    | ...     |
| 2 | Has the data been properly formatted?         | ...     |
| 3 | What trends and relationships were found?     | ...     |
| 4 | What insights help answer business questions? | ...     |

## 5. Report

### 5.1 Key tasks and deliverable(s)

5.1.1. Determining communication approach

5.1.2. Data visualizations creating

5.1.3. Presenting findings

5.1.4. Ensuring accessibility

5.1.5. Support visualizations and key findings

### 5.2 Guiding questions

| N | Questions                                  | Answers |
|---|--------------------------------------------|---------|
| 1 | Were the business questions answered?      | ...     |
| 2 | What story does the data tell?             | ...     |
| 3 | Who is the audience                        | ...     |
| 4 | Is visualization effective and accessible? |         |

## 6. Usage results

### 6.1. Key tasks and deliverable(s)

#### 6.1.1. Top three recommendations based on analysis

### 6.2 Guiding questions

| N | Questions                                | Answers |
|---|------------------------------------------|---------|
| 1 | What is the final conclusion?            | ...     |
| 2 | How can the business apply the insights? | ...     |
| 3 | What are the next steps?                 | ...     |
| 4 | Is additional data needed?               | ...     |